Data Cleaning Log

Step-by-step record of the transformations, filters, checks, and row counts captured in the supplied cleaning log. Row counts and notes are reproduced from the CSV.

Cleaning step	Rows before	Rows after	Rows affected	Notes / rule
NOAA date: YYYYMMDD number to date	137,187	137,187	137,187	0 failed (a bad date would raise an error)
Open-Meteo time: text to datetime	105,192	105,192	105,192	0 failed
Number strings check	137,187	137,187	0	NOAA value is int64, forecasts are float64: no numbers stored as text
Spelling and case (city, element)	137,187	137,187	0	0 values changed
Units: tenths to deg C and mm	4,535	4,535	4,535	value / 10, rows converted
Station and grid point are the same place	3	3	0	rule: under 10 km, 3 of 3 matched
NOAA: keep project window	137,187	10,385	126,802	2022-06-01 to 2025-08-31
Open-Meteo: keep project window	105,192	85,608	19,584	one extra day before start
NOAA: drop rows failing quality check	10,385	10,385	0	q_flag not blank
NOAA: keep TMAX and PRCP	10,385	4,535	5,850	other elements not used
NOAA: drop -9999 codes	4,535	4,535	0	GHCN missing code
NOAA: drop exact duplicates	4,535	4,535	0	rule: all columns equal
NOAA: drop duplicate key	4,535	4,535	0	rule: key city+date+element
Open-Meteo: drop exact duplicates	85,608	85,608	0	rule: all columns equal
Open-Meteo: drop duplicate key	85,608	85,608	0	rule: key city+time
Rebuild full calendar	3,145	3,564	419	gap days added as NaN
TMAX table: cut days after end of station record	3,564	3,542	22	station record stops before the window end
Rain table: cut days after end of station record	1,770	1,743	27	station record stops before the window end
TMAX table: drop days with any missing forecast lead	3,542	3,461	81	same days kept for all 7 leads, API gap, not imputed
Rain table: drop days with any missing forecast lead	1,743	1,722	21	same days kept for all 7 leads, API gap, not imputed
Missing-value test, PRCP	1,722	1,722	0	logistic regression, LR p = 8.308e-107

Cleaning step	Rows before	Rows after	Rows affected	Notes / rule
Missing-value test, TMAX	3,461	3,461	0	logistic regression, LR p = 7.378e-33
TMAX: delete days without station value	3,461	2,788	673	no imputation, station TMAX is the truth
Rain: delete days without station value	1,722	822	900	no imputation, station PRCP is the truth
TMAX outliers: delete errors, keep real	2,788	2,787	1	MAD on month anomaly, real values kept
Rain: delete impossible rain (> 500 mm/day)	822	822	0	heavy rain kept, not capped